Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		(annual saving) 12 × 50 = £600 (1) (Total insulation cost) 3 800 + 2 900 = £6 700 (1) Pay back time = 6 700(ecf)/600(ecf) = 11.17 years (1) (so not true) OR (Total insulation cost) 3 800 + 2 900 = £6 700 (1) 6 700 (ecf)/ 50 = 134 months (1) 134 (ecf) /12 = 11.17 years (1) (so not true) If ecf gives less than 10 years – so true (if the ecf calculation is correct only 2 marks 1 mark for correct judgement based on incorrect answer. Accept: Total insulation cost = £3 800 + £2 900 = £6 700 (1) Amount paid back in 10 years = 10 ×12 × 50 = £6 000 (1) £700 not paid off so not true / more needed to pay off cost (1)			3	3	3	
		(ii)		Any 3 × (1) from: Maintenance costs (1) Tilt/angle of panel/not facing the Sun (1) Cost of electricity may decrease (1) less sunshine hours than expected / more cloud (1) Panels are dirty (1) Electricity consumption decreases (1)		3		3		

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
	(b)			14 000W conversion to 14 kW (1) 14 × 1 500(1) = 21 000 kWh (1) 23% × 21 000 = 4 830 kWh (so claim is valid) (1) OR 14 000W conversion to 14 kW (1) 14 × 23 /100 = 3.22 kW (1) 3.22 × 1 500 (1) = 4 830 kWh (so claim is valid) (1)			4	4	4	
	(c)	(i)		3.5×10^6 = 3 500 000 kW (1) 3 500 000 / 4 420 = 791.9 homes (1) Accept any answer 791-792		2		2	2	
		(ii)		$5.3 \times 10^{10} / 3.5 \times 10^6$ (1) = 15 142.857 (1) Accept any answer 15 142 – 15 143		2		2	2	
				Question 4 total		**7**	**7**	**14**	**11**	